

Synthesizing a novel genetic sequential logic circuit: a push-on push-off switch

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Chen^{1,3}, Xiaoyue Chen^{1,3}, Le Yang^{1,2}, Haisu Ma^{1,3}, Jianguo Chen³ & Qi Ouyang^{2,4}

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*Presented by Sana Wajid
November 1, 2011*

Overview

Introduction

Combinational and Sequential Logic

Using synthetic biology for information processing

Design of artificial genetic circuits

Genetic Sequential Logic Circuit (GSLC)

GSLC: Memory Module

Methods Overview

Results Overview

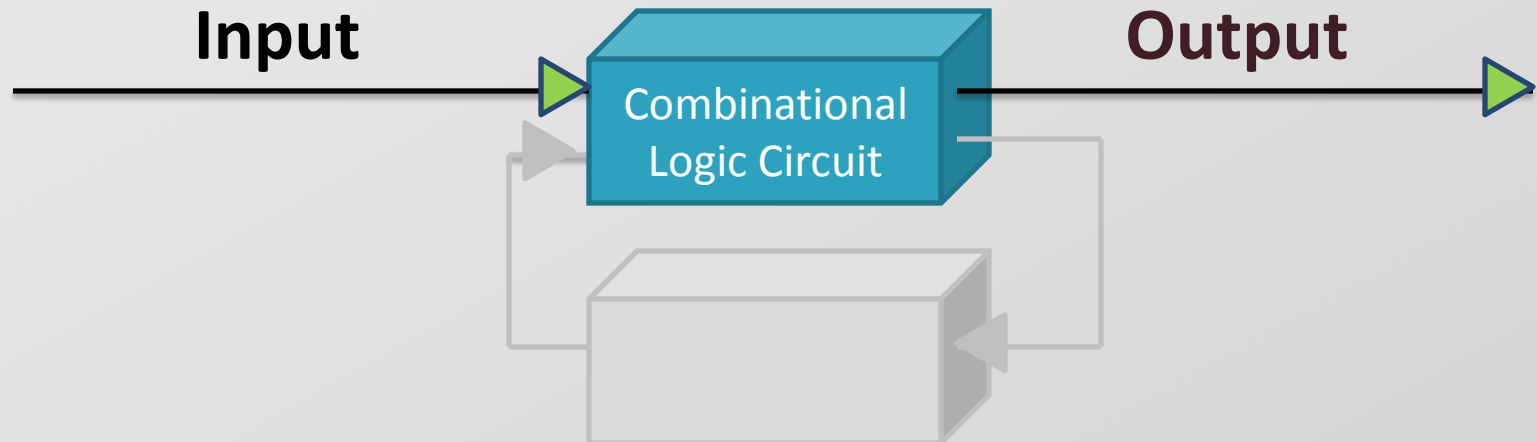
Discussion of main points

References

Combinational and Sequential Logic

Combinational Logic Circuits:

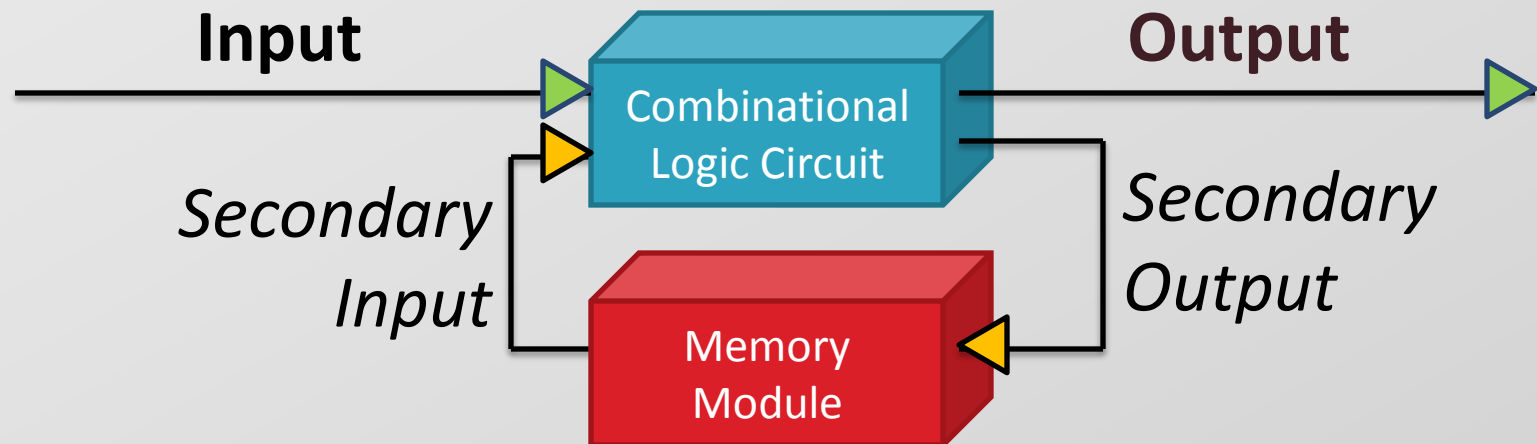
Output depends only on present input value



Combinational and Sequential Logic

Sequential Logic Circuits:

Output depends on the history of inputs,
Past input value and the present input value



*Combinational Logic Circuit connects to memory module
to form a feedback path*

Combinational logic circuit

Input pitcher is variable [empty, full]

Output pitcher is always starts empty (*resets*)



Input



Output

Combinational logic circuit

Input

Output



Input



Output

able [empty, full]

ways starts empty (*resets*)

Combinational logic circuit

Proof: Output pitcher is now strictly dependent on *input* pitcher



Input



Output

Sequential logic circuit

Input pitcher is variable [empty, full]

Output pitcher is carried over (*memory*)



Input



Output

Sequential logic circuit

Input

Output



Input



Output

Carried over (memory)

Carried over (memory)

Sequential logic circuit

Input pitcher is variable [empty, full]

Output pitcher is carried over (*memory*)



Input



Output

Synthetic Biology

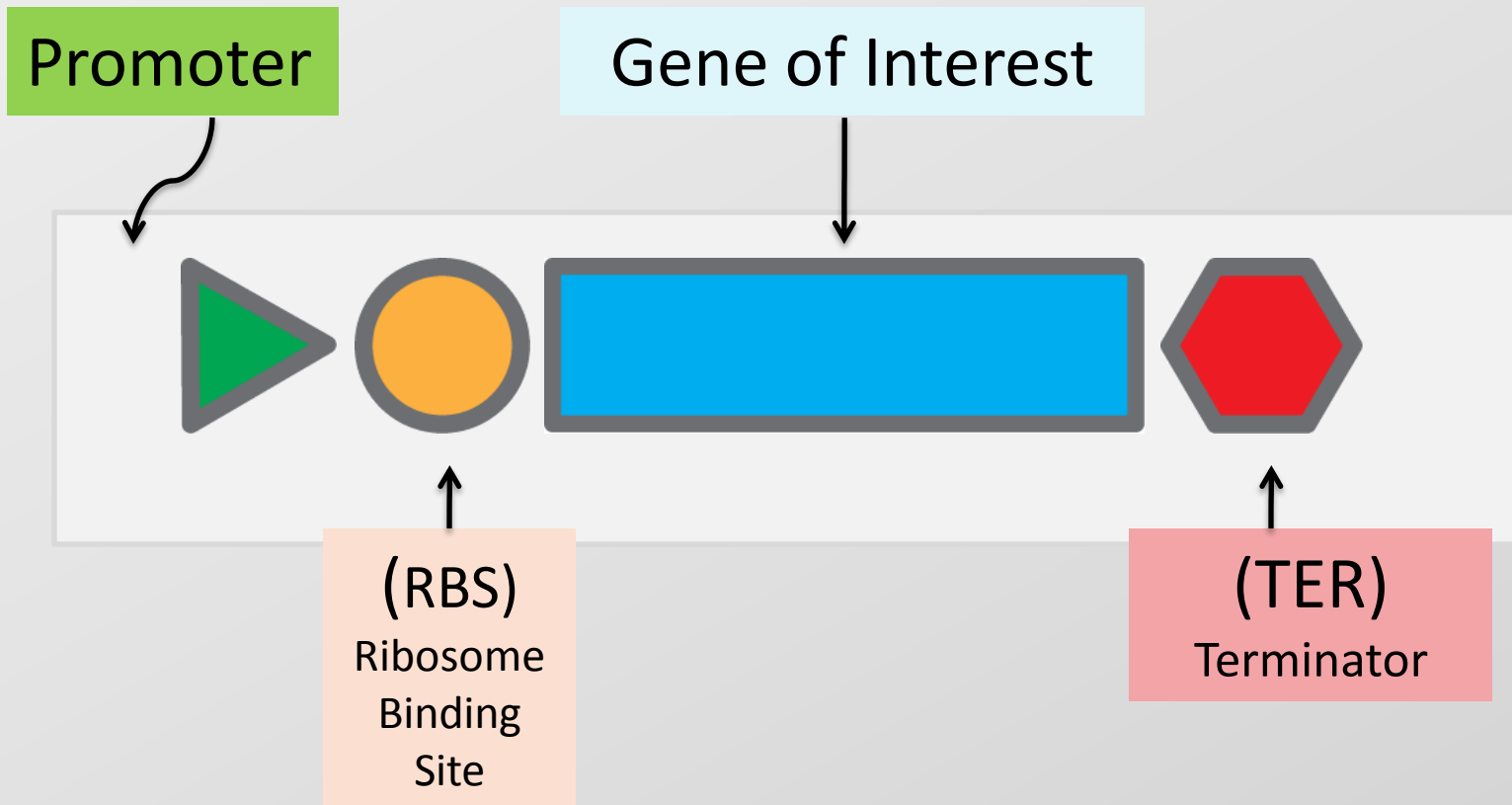
Synthetic Biology

Emulating the concepts in engineering such as logic circuits, in biology

Can combinational logic and sequential logic occur in biology, naturally or synthetically?

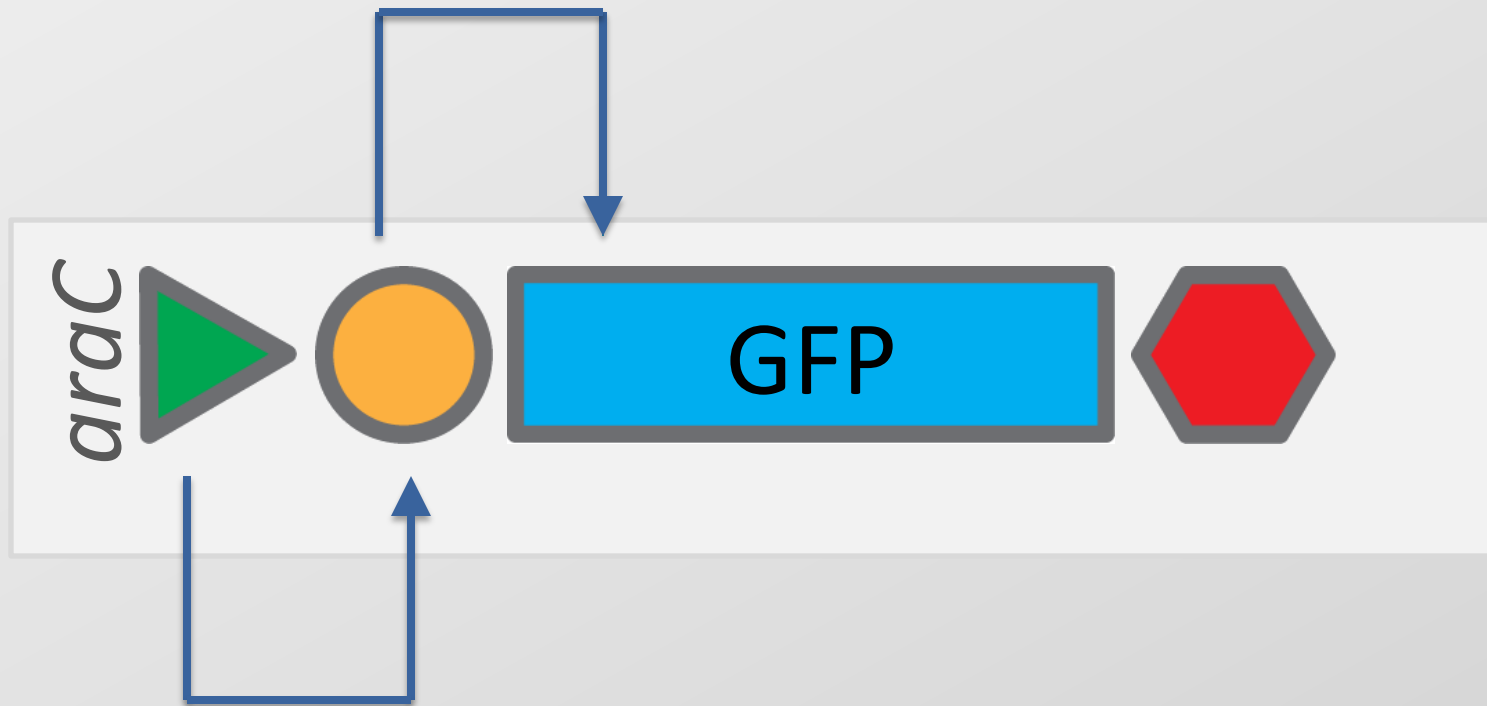
Design of artificial genetic circuits

Graphical abstraction



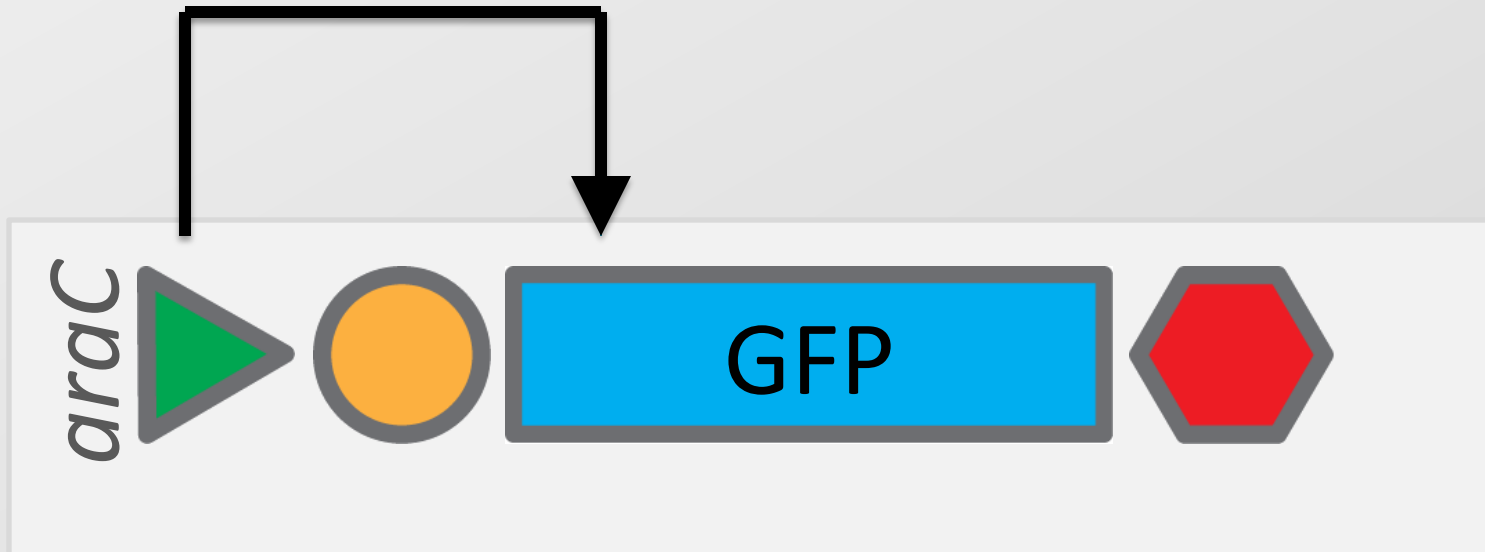
Design of artificial genetic circuits

Genetic combinational logic circuit



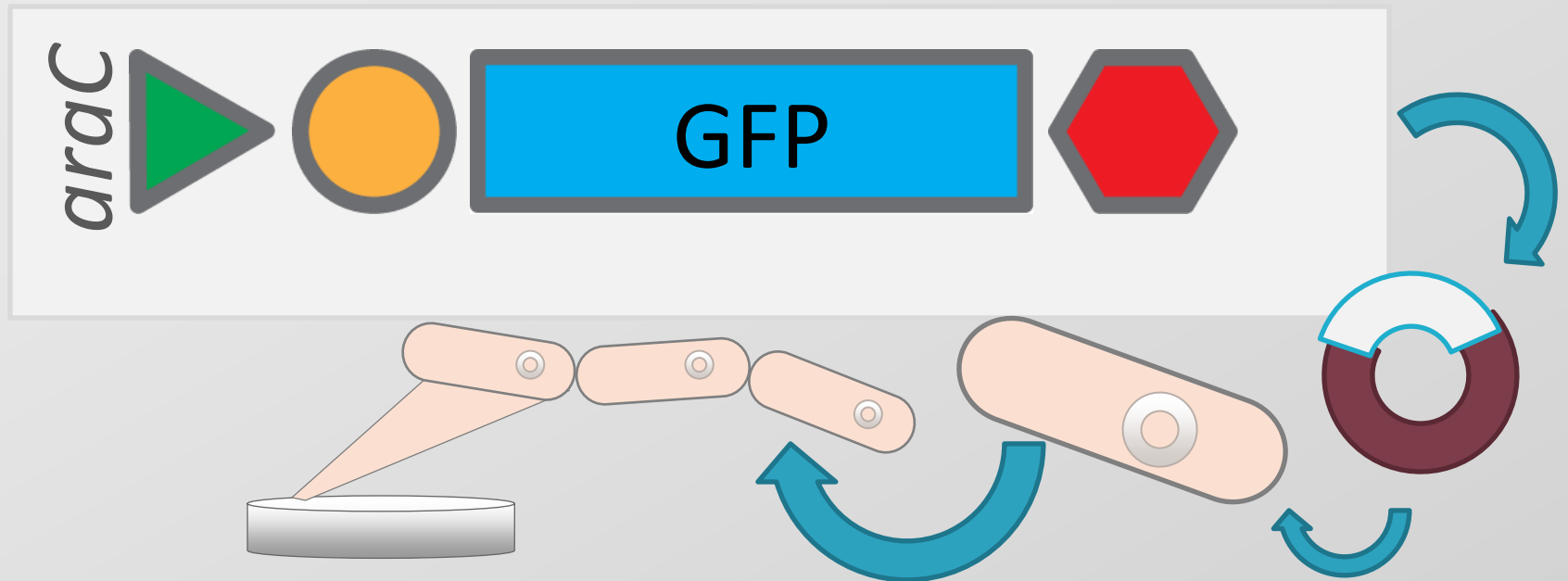
Design of artificial genetic circuits

Genetic combinational logic circuit



Design of artificial genetic circuits

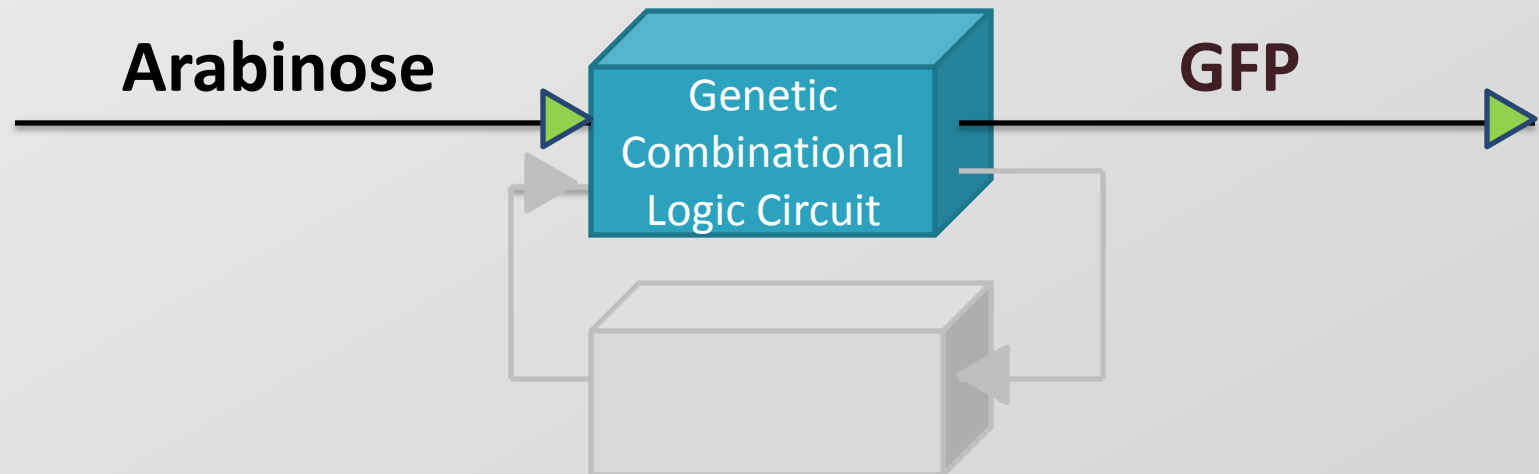
Genetic combinational logic circuit



Design of artificial genetic circuits

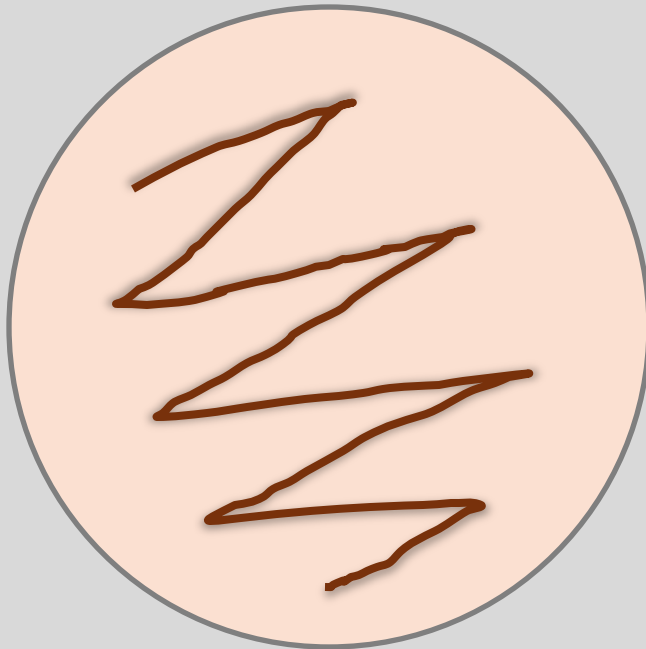
Genetic Combinational Logic Circuits (GCLC):

Output depends only on present input value

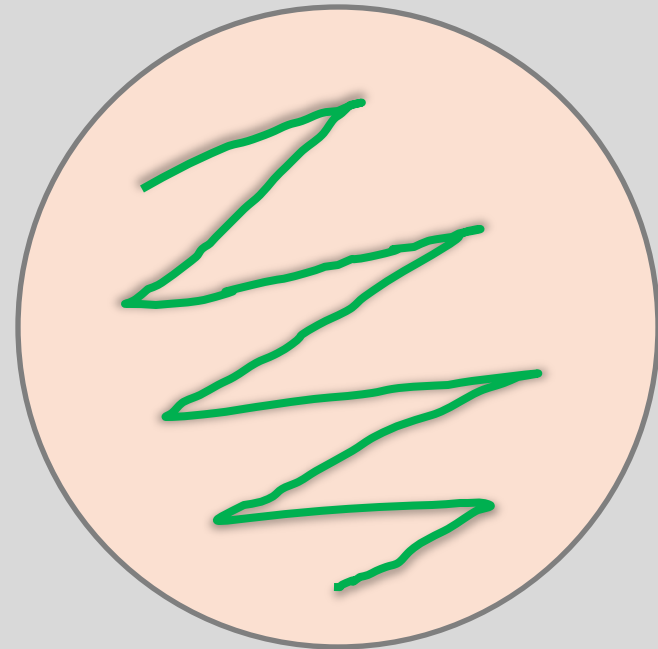


Design of artificial genetic circuits

Proof of concept



Medium lacking arabinose,
No GFP expression



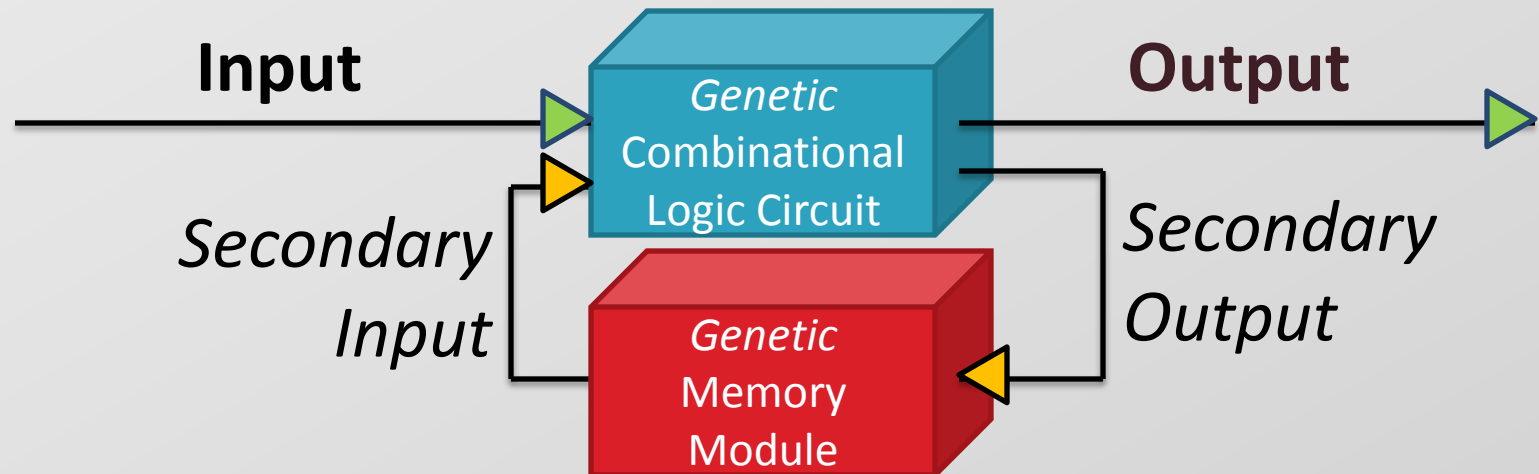
Medium supplemented with arabinose
GFP expression is observed

Genetic combinational logic circuit consistent with digital combinational logic circuit!

Combinational and Sequential Logic

Genetic Combinational Logic Circuits (GSLC):

Output depends on the history of inputs,
Past input value and the present input value



*Combinational Logic Circuit connects to memory module
to form a feedback path*

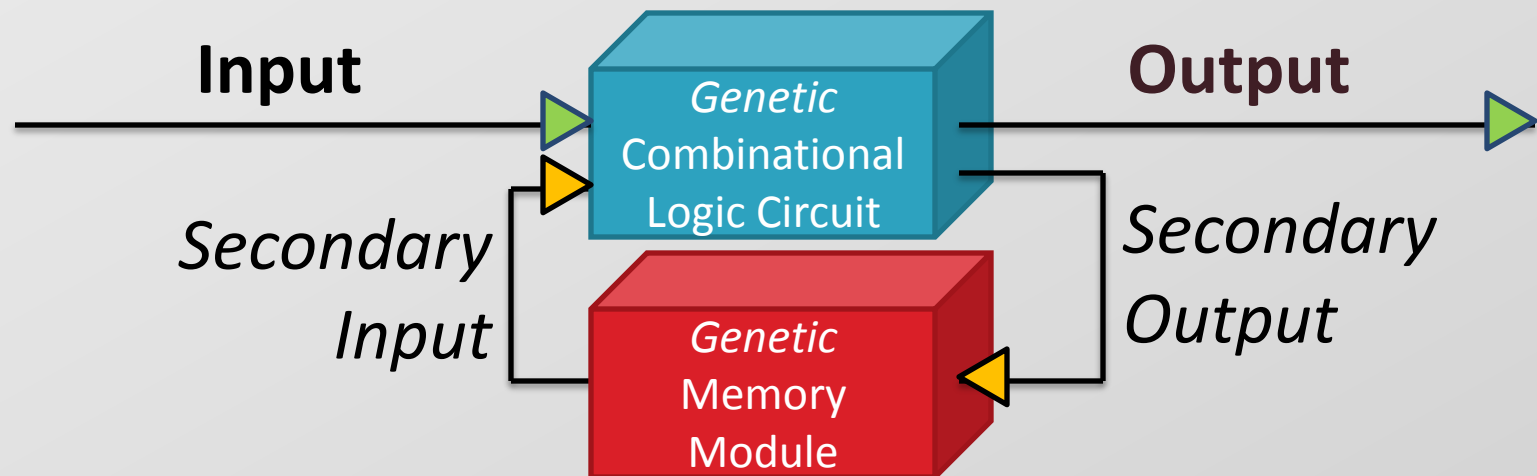
Genetic Sequential Logic Circuit (GSLC)

Peking 2007 iGEM team's novel GSLC consists of:

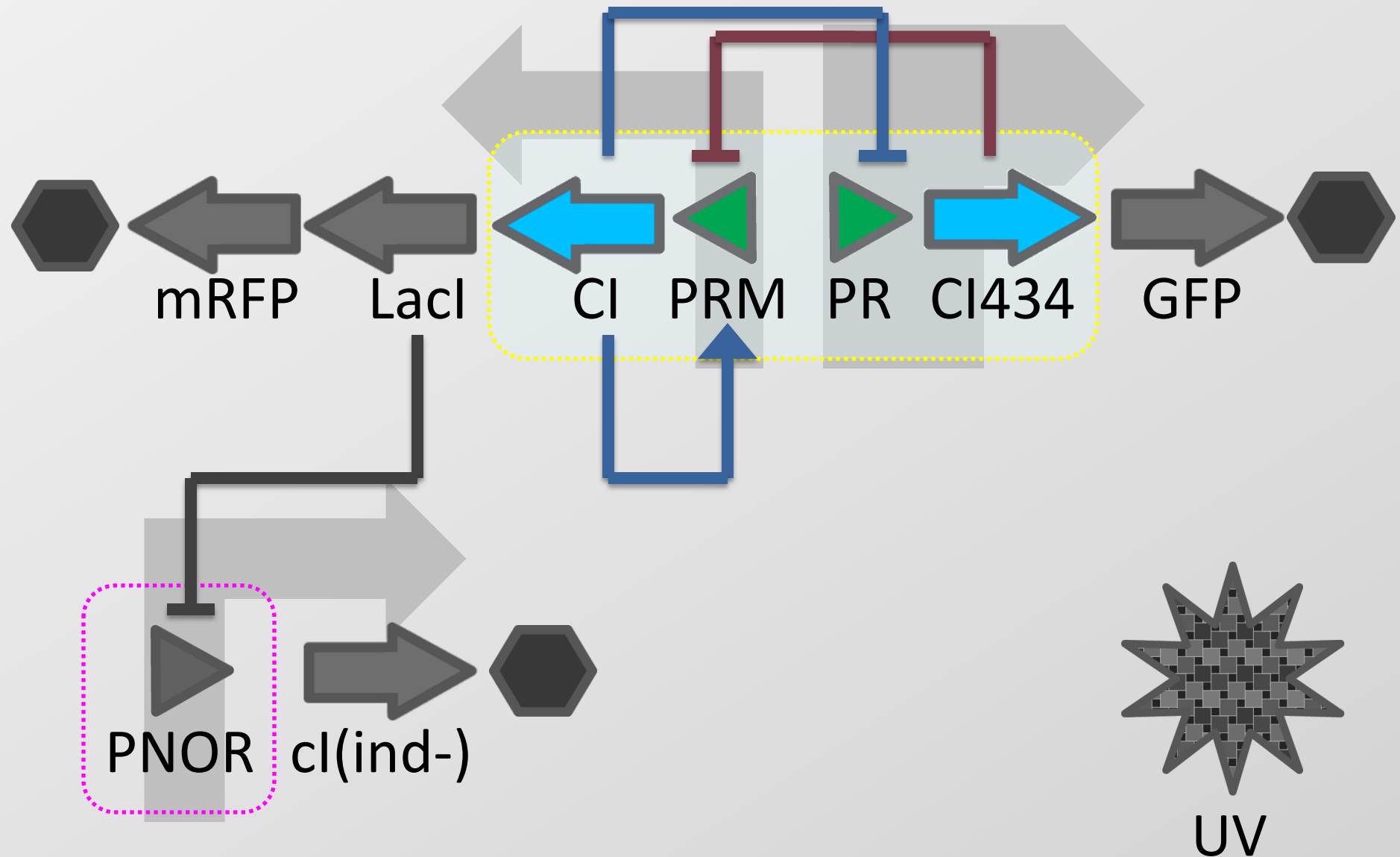
- I. A bi-stable switch memory module
- II. A double-repressed promoter
NOR gate module
- III. Two interconnecting parts connecting the
two modules

Genetic Sequential Logic Circuit (GSLC)

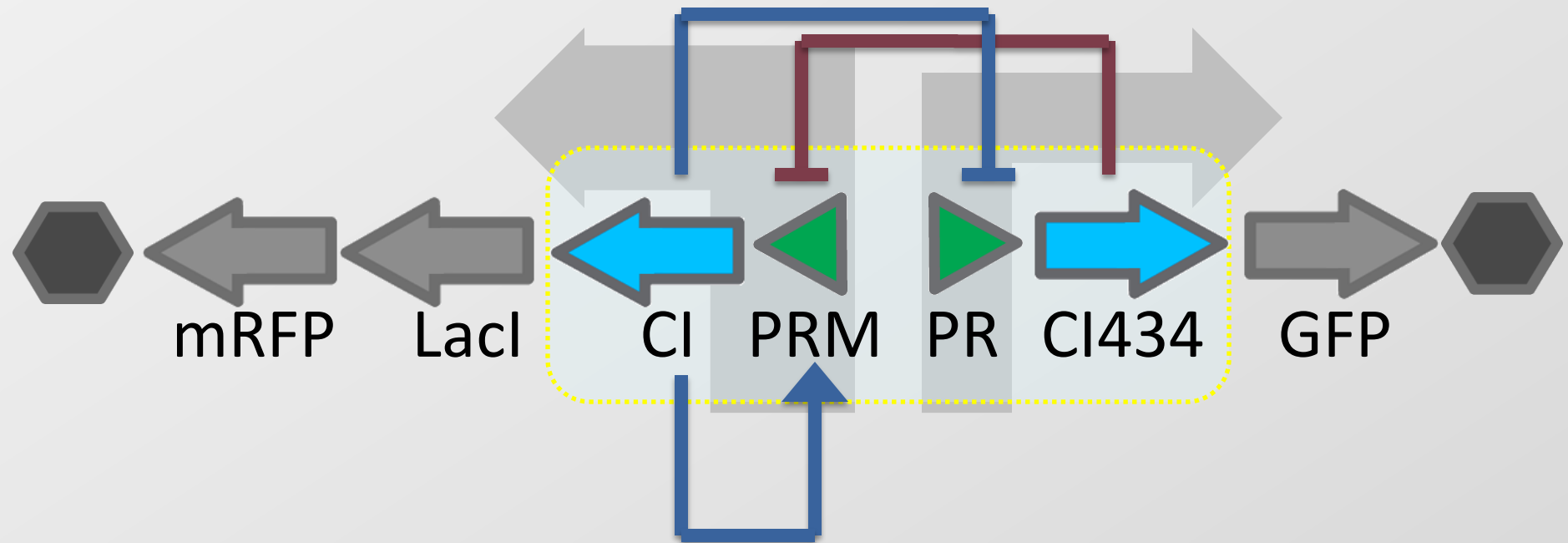
- ✓ Generates output based on: a present input signal and an internal state
- ✓ Output feeds back to the memory module such that the **history** of the output signal can be registered and thereby saved



GSLC: Memory Module



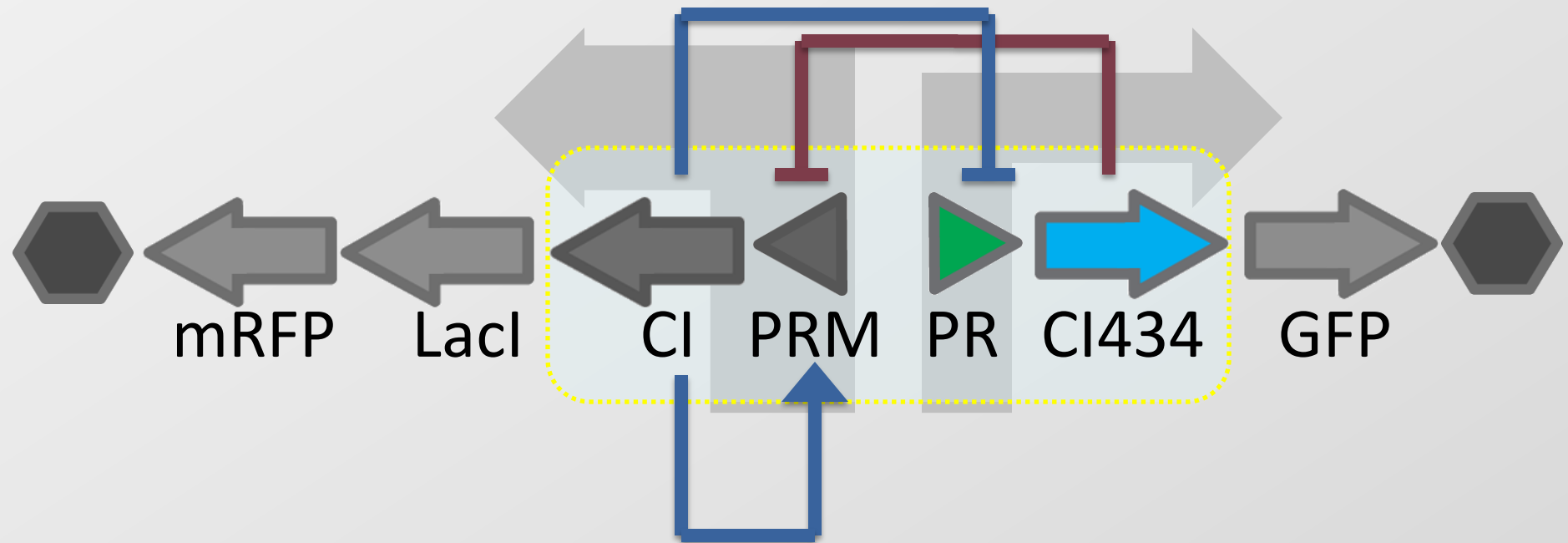
GSLC: Memory Module, OFF State



CI434 is present, it can repress the transcription of *ci*, thus maintaining the transcription of itself, GFP is expressed

Output: low CI/high CI434 state => GFP

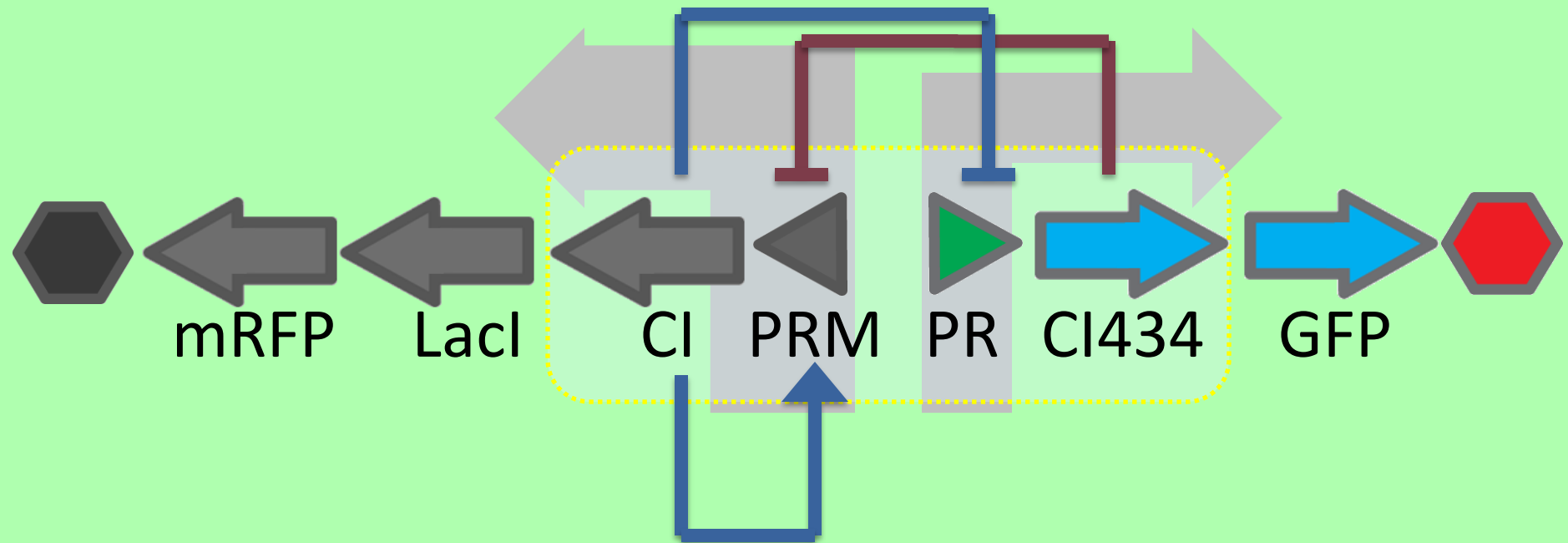
GSLC: Memory Module, OFF State



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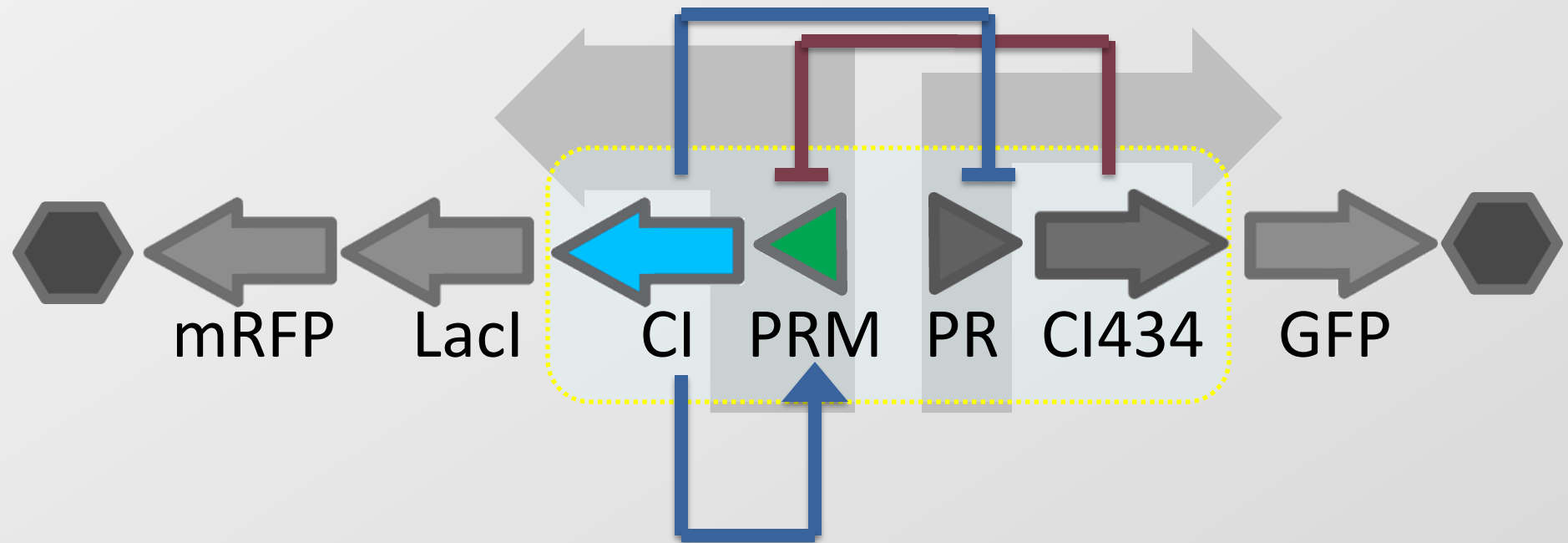
GSLC: Memory Module, OFF State



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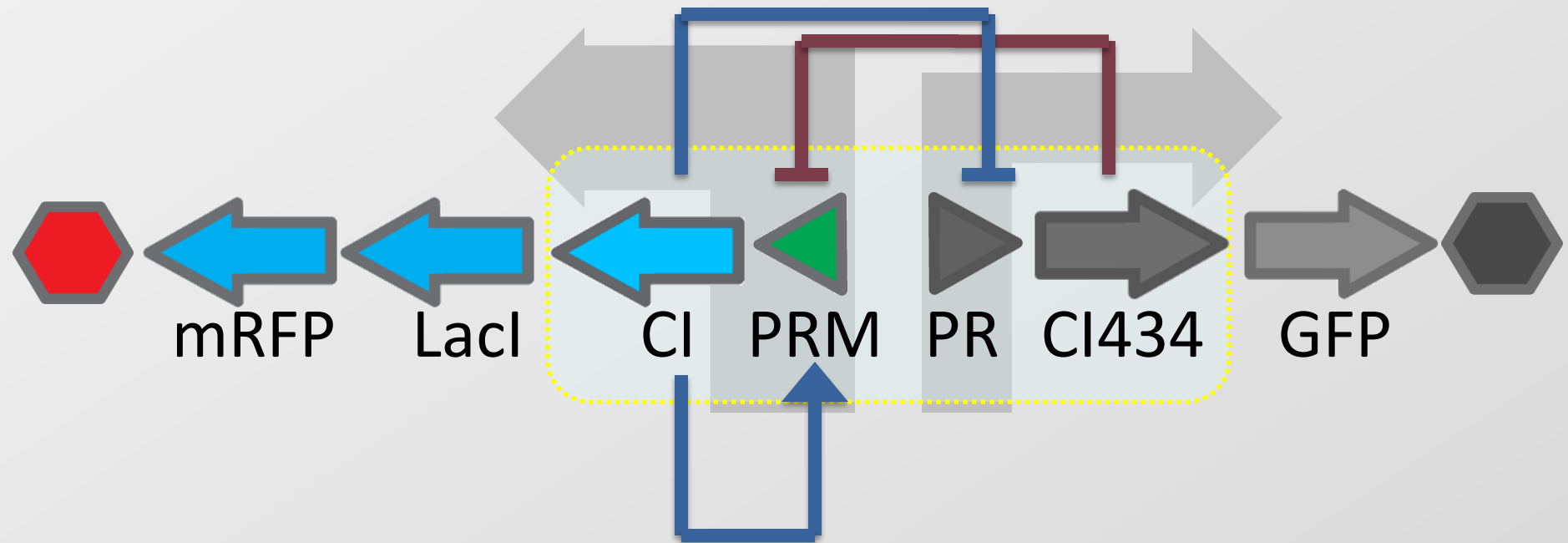
GSLC: Memory Module, ON State



CI is present, it can activate its transcription
and repress the transcription of ci434,
RFP is expressed

Output: high CI/low CI434 state => RFP

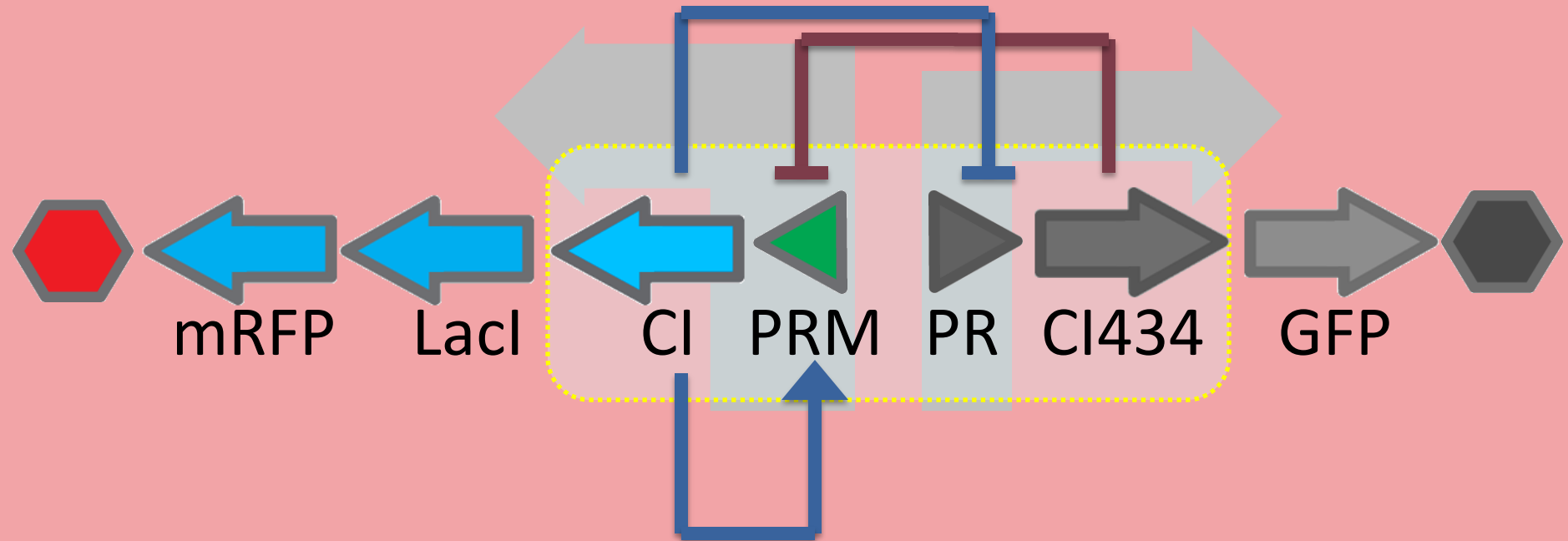
GSLC: Memory Module, ON State



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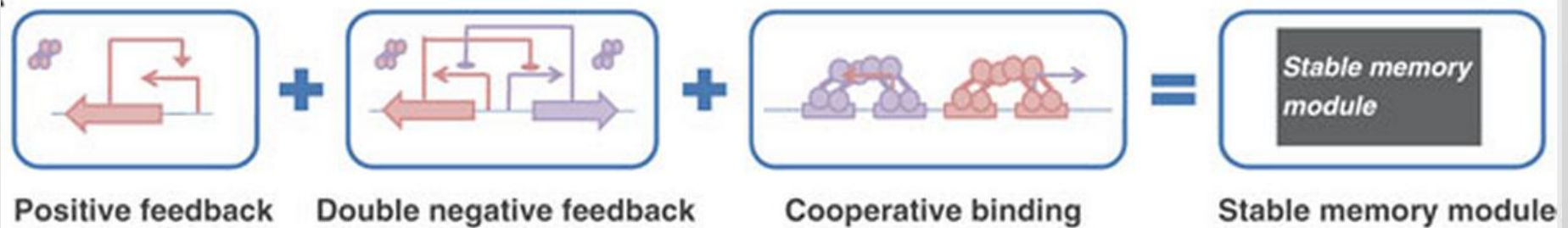
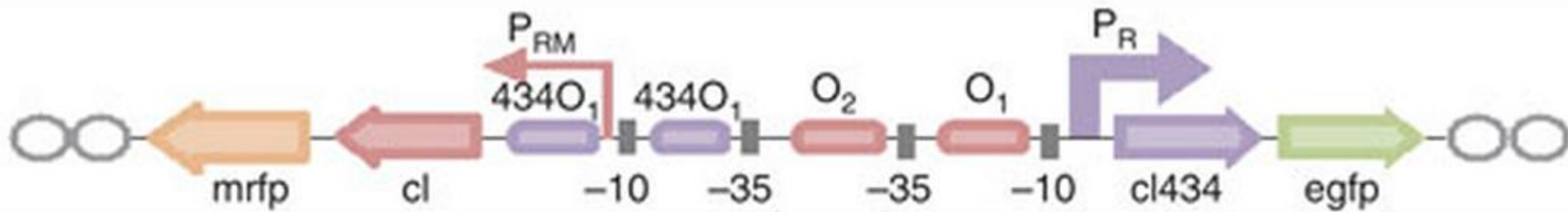
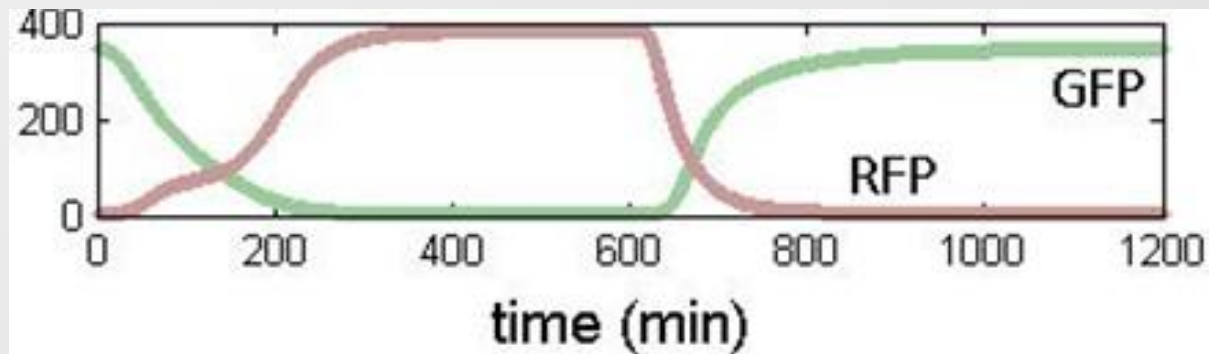
GSLC: Memory Module, ON State



CI is present, it can activate its transcription
and repress the transcription of ci434,
RFP is expressed

Output: high CI/low CI434 state => RFP

GSLC: Memory Module



Example: Cl dimers cooperatively bind O₁ and O₂
-> activate P_{RM} & repress P_R

GSLC: Nor gate Module

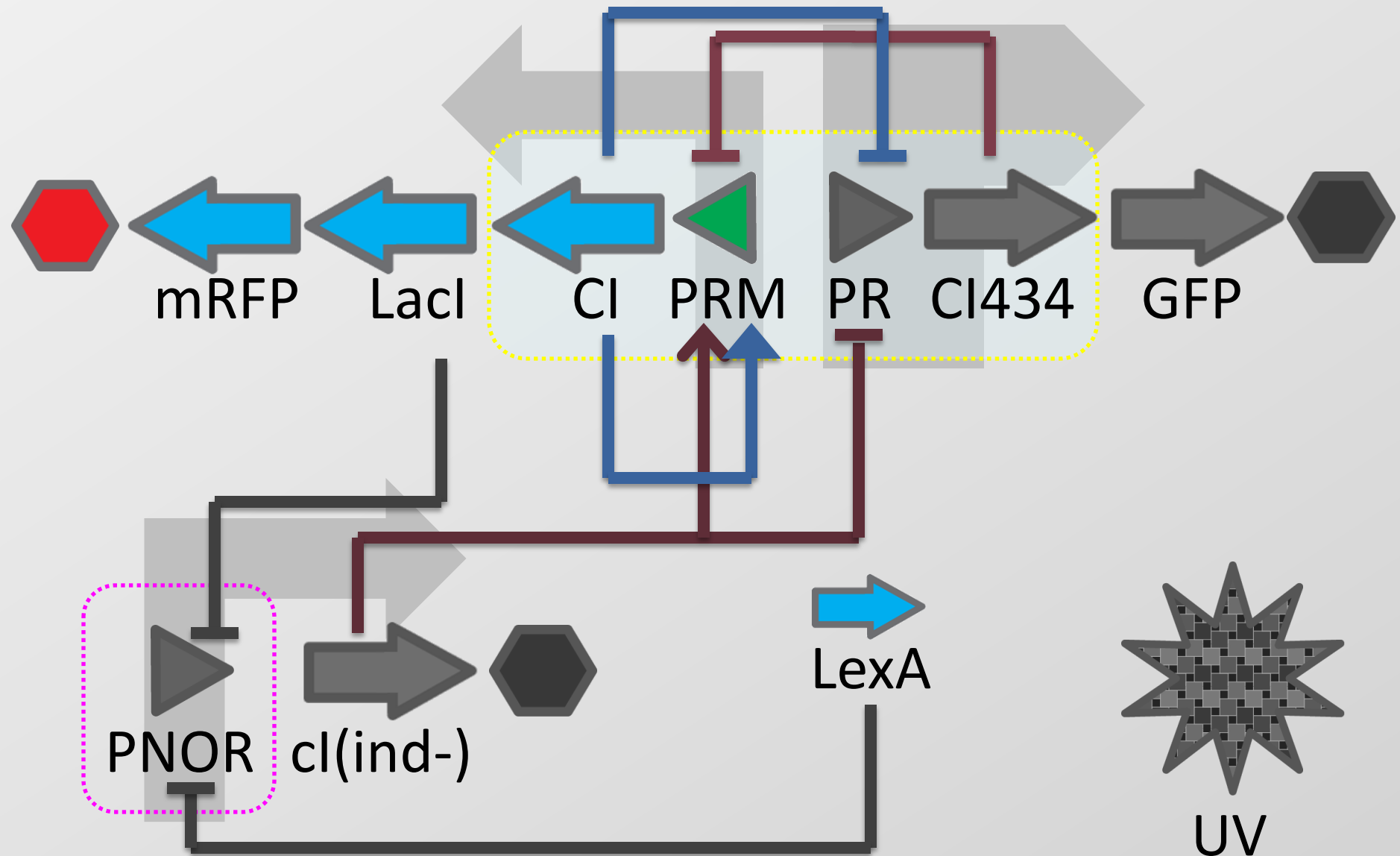
A NOR Gate will only equal true when nothing is added to the system (A & B are false)

PNOR is repressed by LacI or LexA

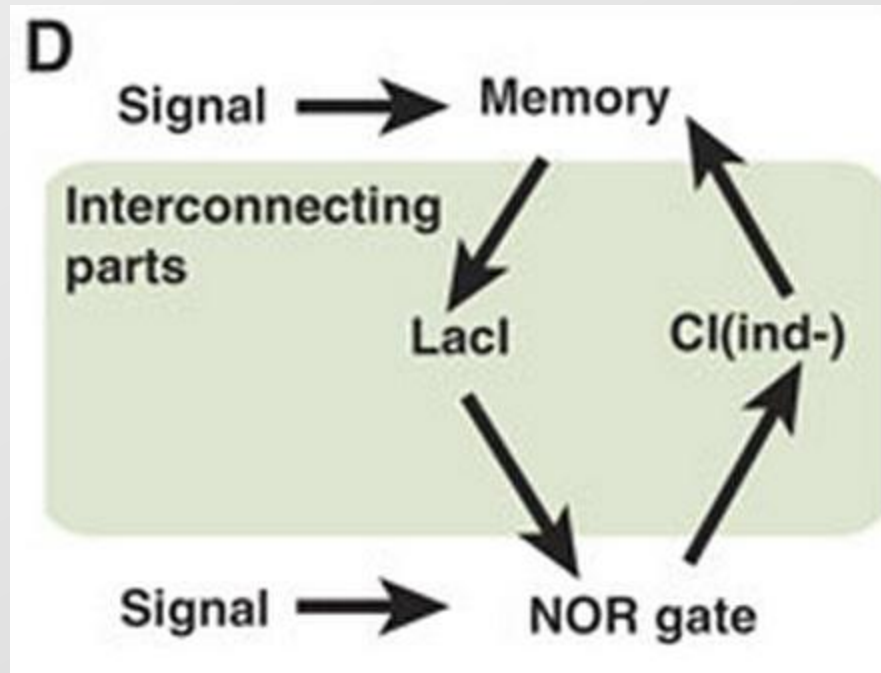
PNOR ON State => both LacI and LexA are off

PNOR OFF State => either LacI or LexA are on

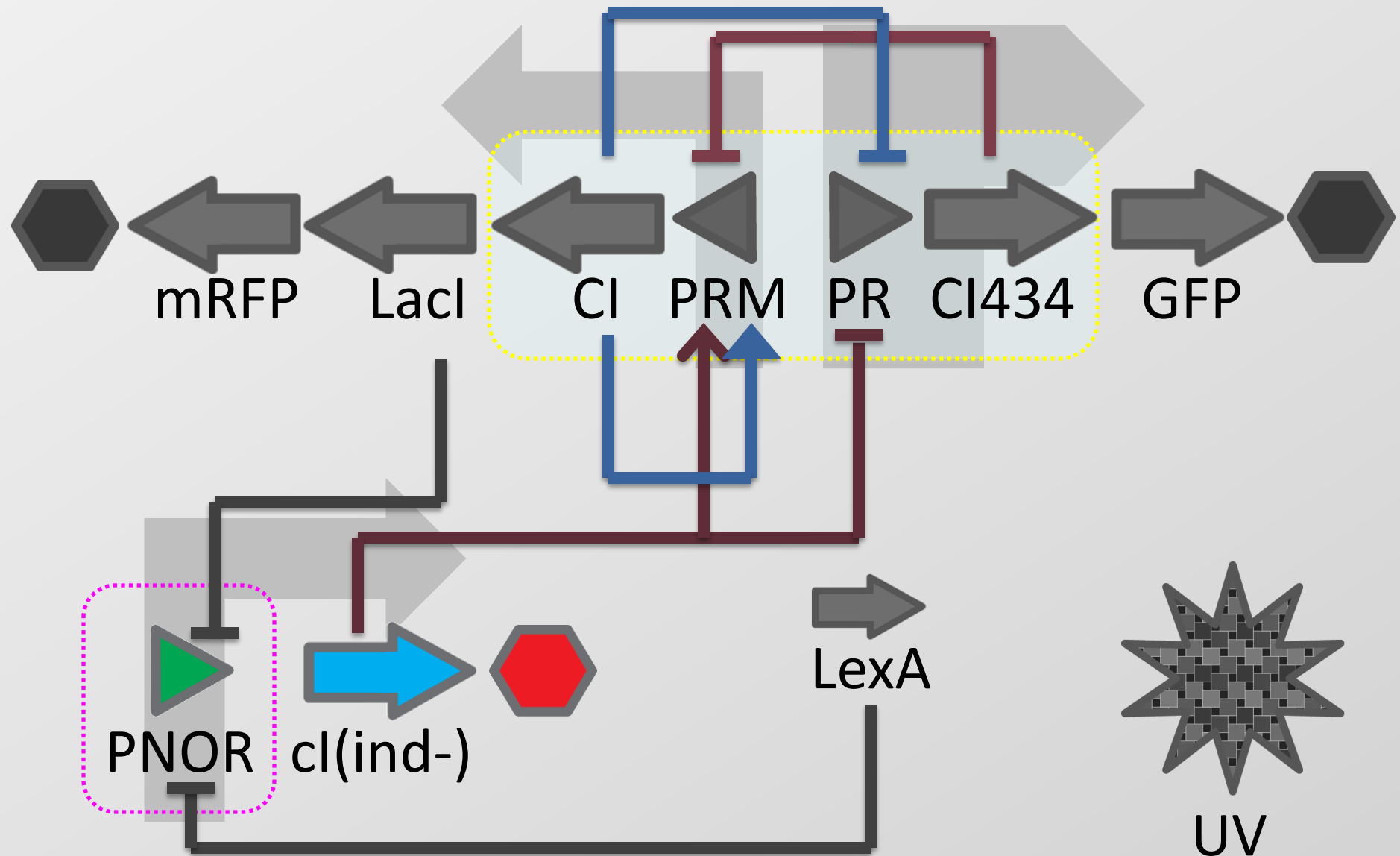
GSLC: Nor gate Module, OFF State



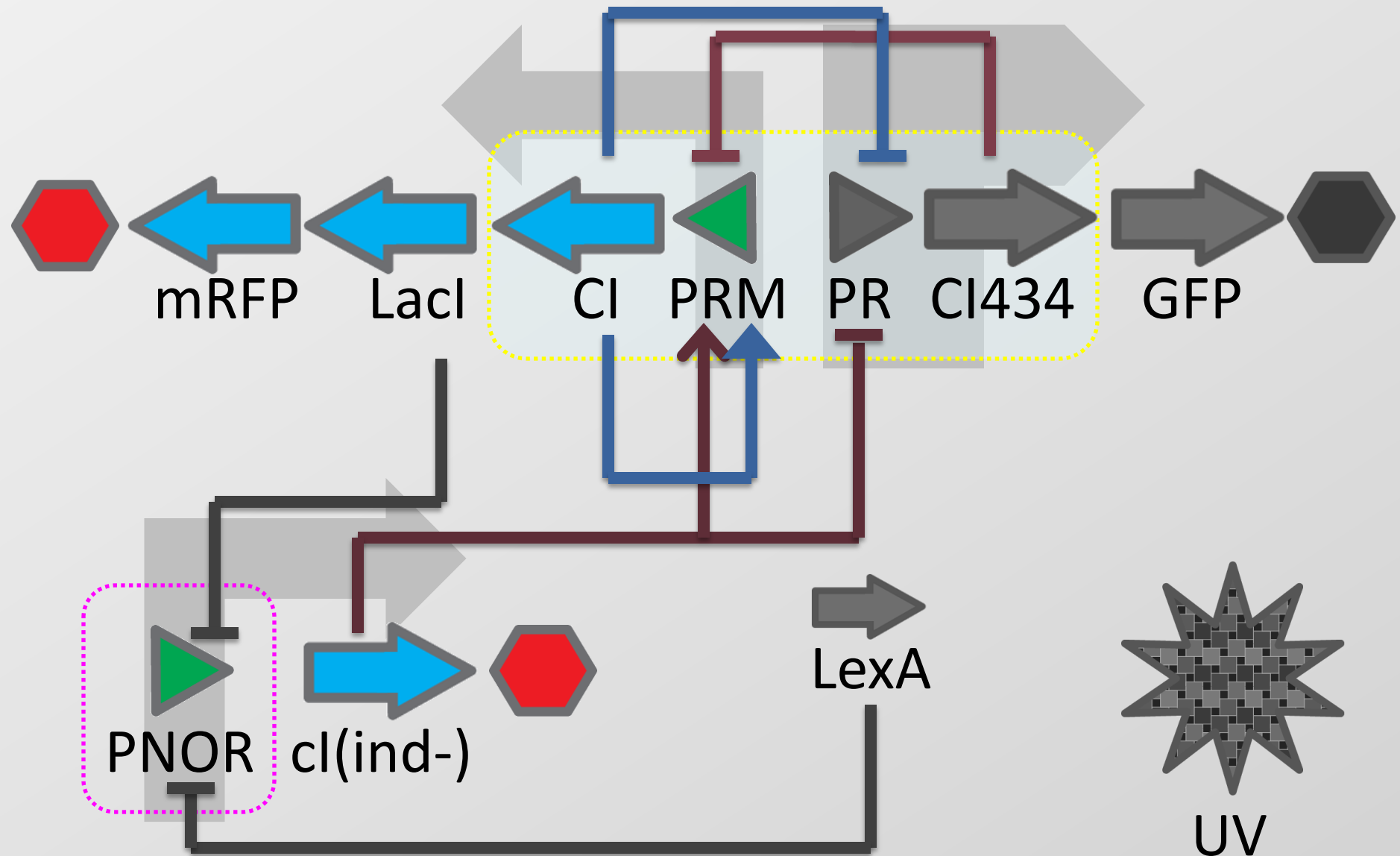
GSLC: Interconnecting Parts



GSLC: Nor gate Module, ON State

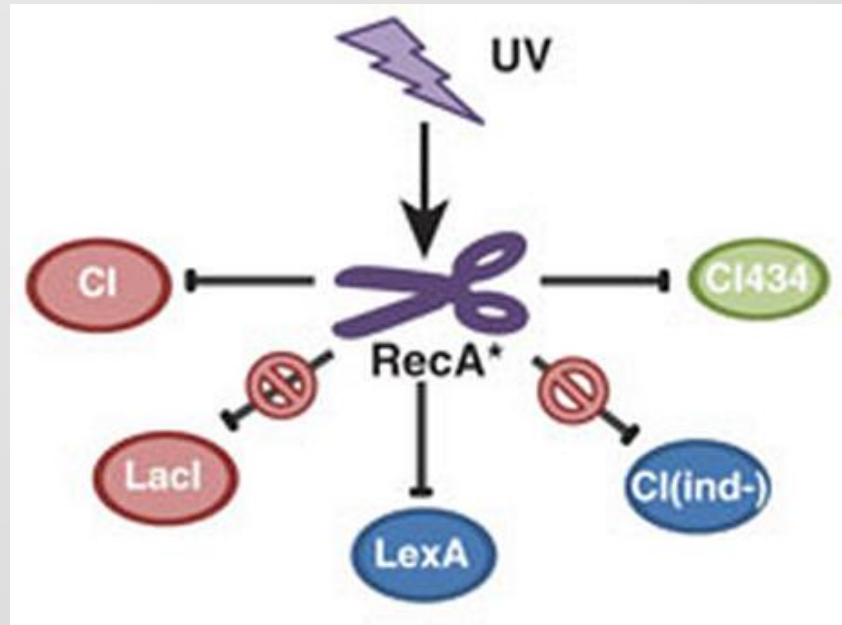


GSLC: Nor gate Module, ON State



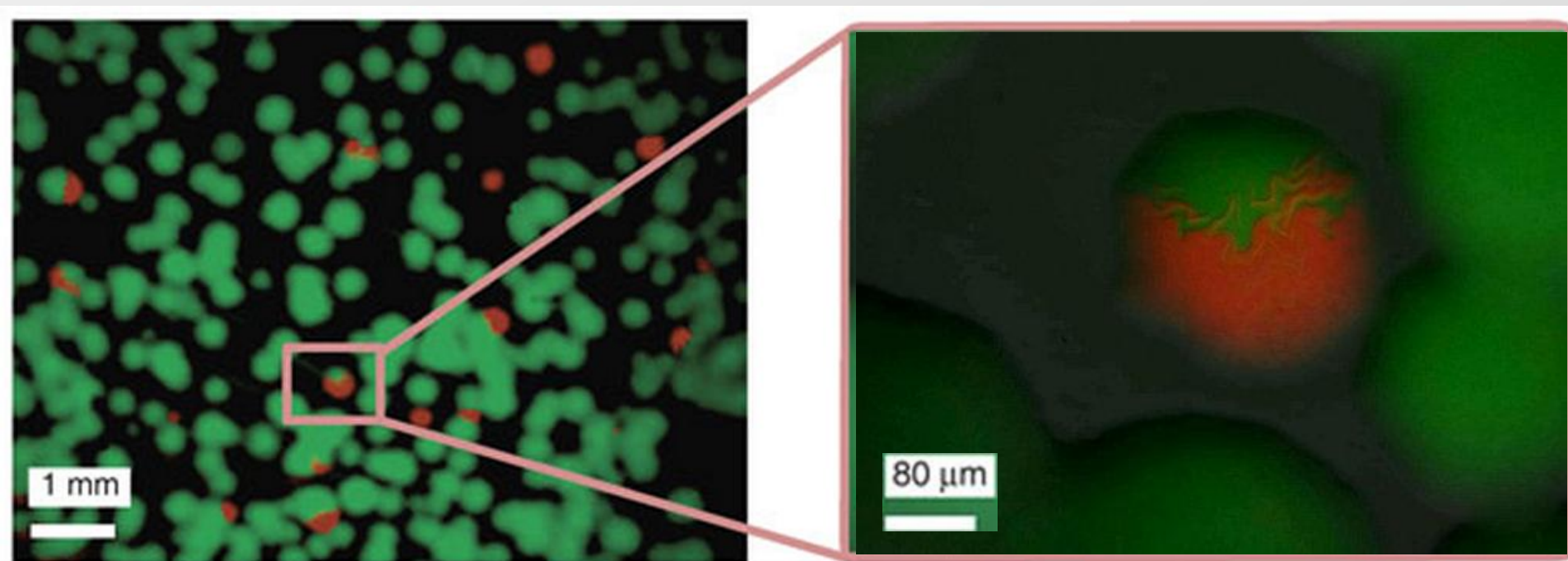
Reset Switch through External Input UV

UV will convert *RecA* to *RecA** (protease)
*RecA** will degrade: *CI434*, *CI*, and *LexA*, but not
LacI and *Clind*



Function as a push-on, push-off switch

Images of cells carrying the memory module



Source: Peking 2007 iGEM Wiki

Function as a push-on, push-off switch

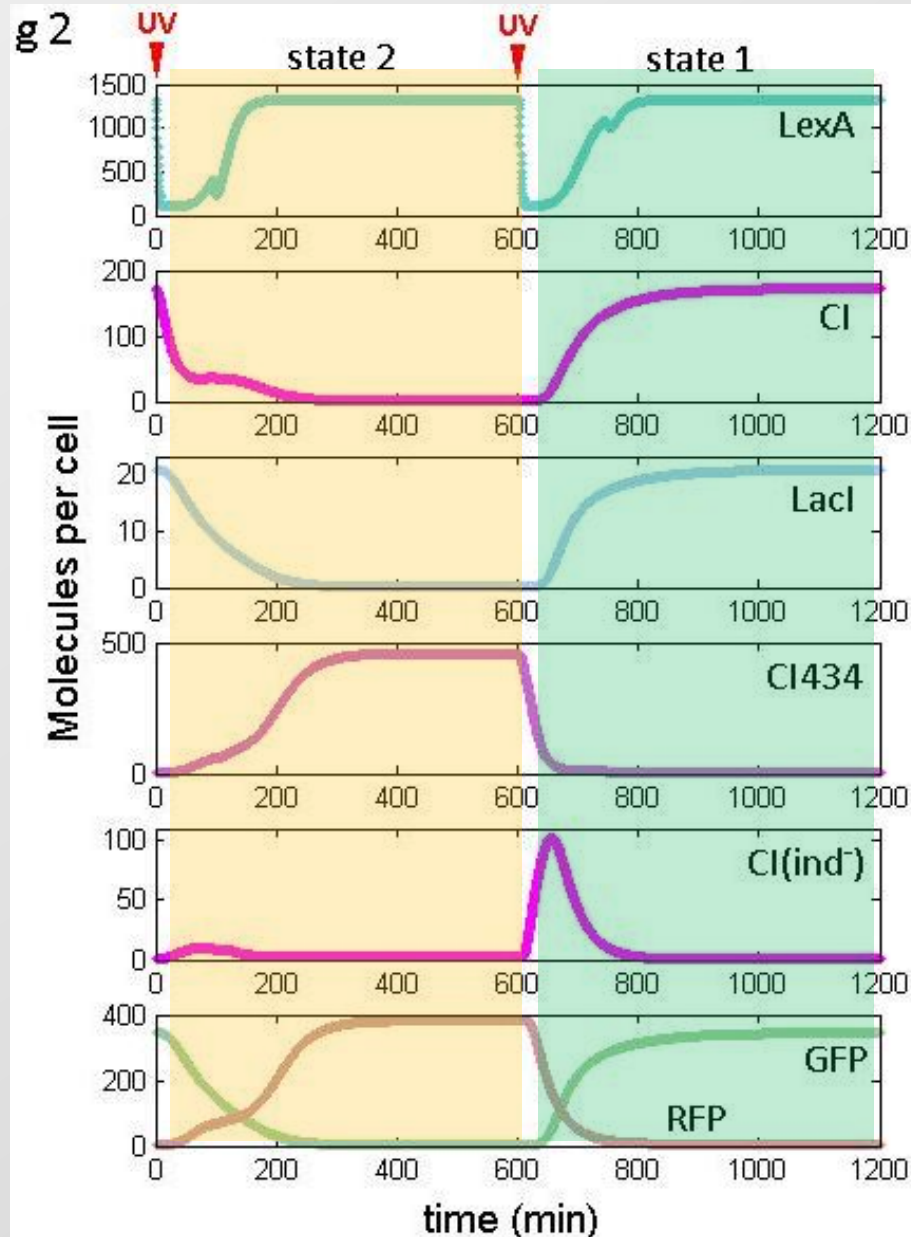
Internal memory is “ON”

the external UV input makes the circuit's output promoter PNOR generate an ‘OFF’ pulse signal and register the ‘OFF’ state into the memory

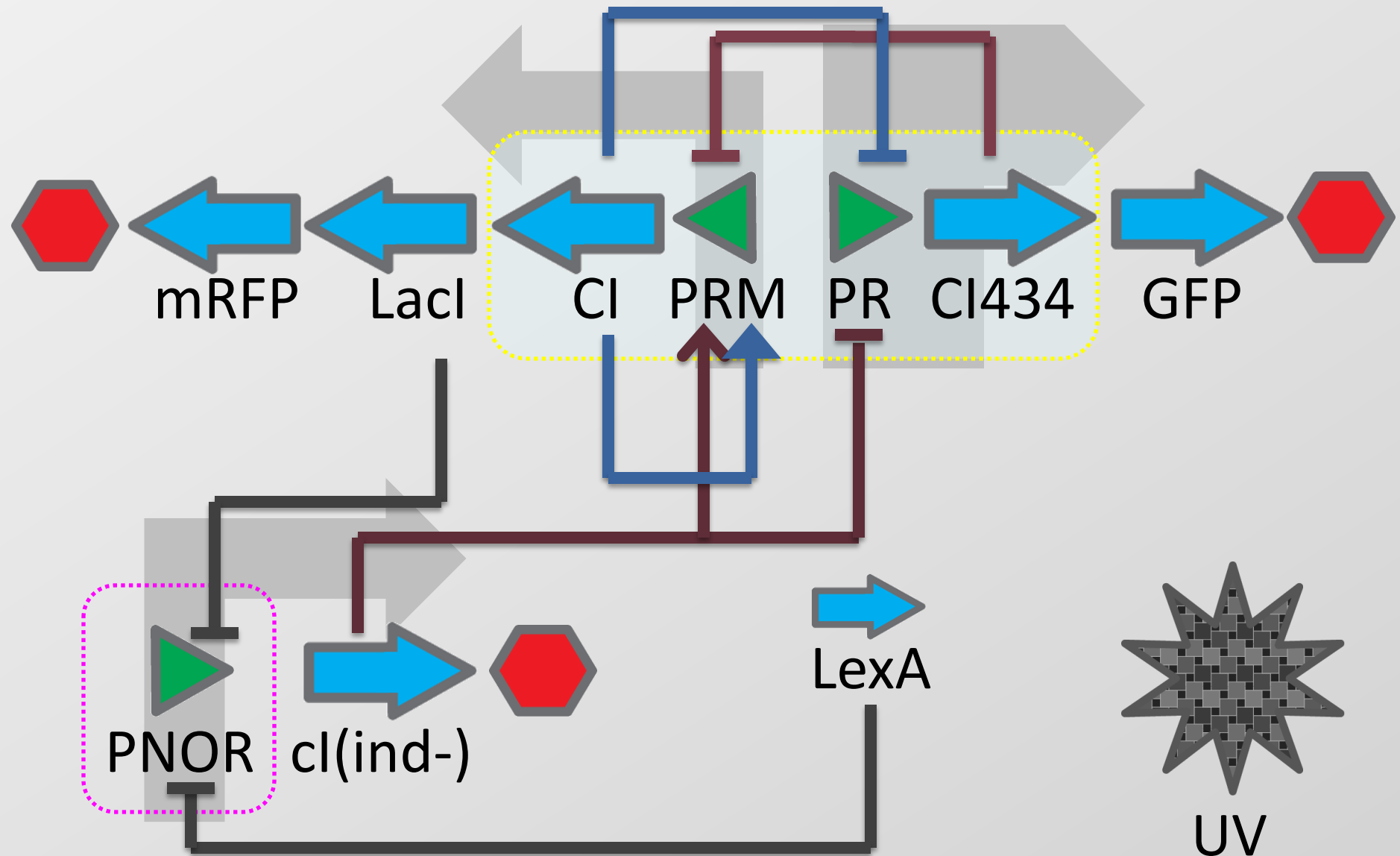
Internal memory is “OFF”

the same external UV input induces the circuit's output promoter PNOR to generate an ‘ON’ pulse signal and register the ‘ON’ state into the memory.

Quant. measurement of GSLC switching

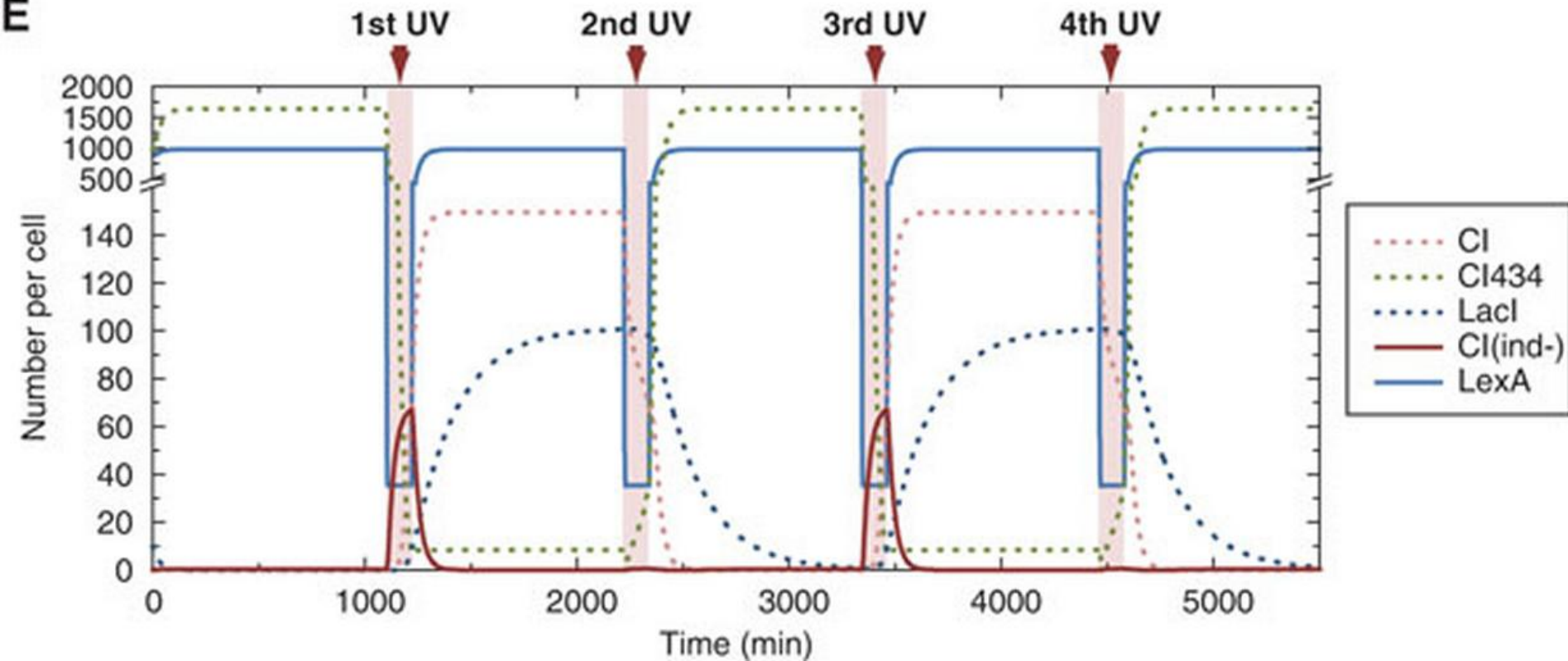


GSLC: Nor gate Module, ON State



Quant. measurement of GSLC switching

E



Methods Overview

Directed evolution was used to select proper coupling strengths between the two modules.

This includes, a fine-tuned expression of the two interconnecting parts LacI and Cl_{ind}- by mutating their ribosome-binding sites (RBSs).

	Wild type or mutant of SDA (the RBS of the cI _{ind} -gene)	Wild type or mutant of SDY (the RBS of the lacI gene)
Wild type	5'-AGGAGGAAAAAAATG-3'	5'-AGAGGAGAAATTAAGCATG-3'
Mutation library	5'-ANNNNNAAAAAAATG-3'	5'-AGNNNNNAAATTAAGCATG-3'
8A	5'-ATGTCGAAAAAAATG-3'	5'-AGTGTGTAAATTAAGCATG-3'
11E	5'-ATGTCGAAAAAAATG-3'	5'-AGGTTTTAAATTAAGCATG-3'
11H	5'-ATGTGTAAAAAAATG-3'	5'-AGTGAGTAAATTAAGCATG-3'

RBS, ribosome-binding site.

The mutated nucleotides and start codons are in bold. N=A, T, C, or G.

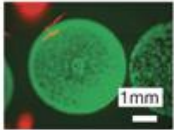
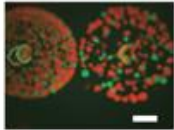
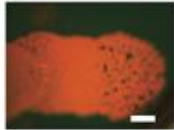
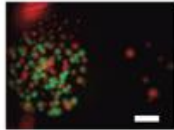
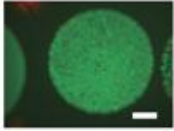
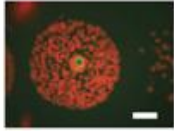
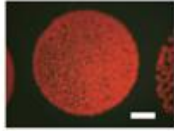
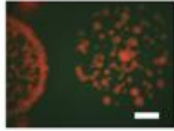
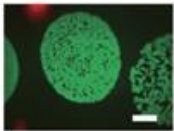
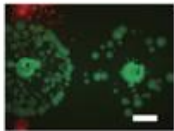
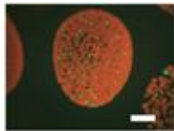
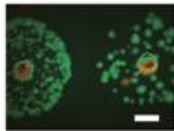
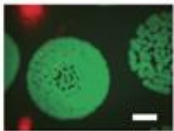
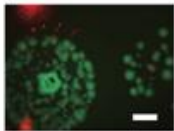
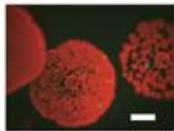
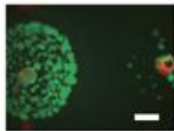
Results Overview

- Switch from 'OFF' to 'ON' is quite complete.
- 'ON' to 'OFF' was not as efficient: only about one-third of the population switched to the 'OFF' state after UV stimulation
 - Reasons: leaky Ci_{ind-} expression, incomplete degradation of CI , insufficient CI_{434} repression accumulation

Control Circuits:

- In the other two control circuits, CI_{ind-} , or both $LacI$ and CI_{ind-} , was removed.

Results Overview

8A Mutant		Green → Red		Red → Green	
LacI	CI(ind-)	Null UV-triggered	UV-triggered	Null UV-triggered	UV-triggered
+	+				
Control		Green → Red		Red → Green	
LacI	CI(ind-)	Null UV-triggered	UV-triggered	Null UV-triggered	UV-triggered
-	+				
+	-				
-	-				

Results Overview

- Sectored colonies during growth in *E. coli*
 - during growth some competent cells with the newly plasmid did not establish a stable inheritable state
 - each of the newborn cells entered different stable states
 - These were inherited to their offspring
 - *Switch is bistable and inheritable!*

Discussion of main points

GSLC can generate different output signals under the same input (UV)

It relies on its internal state of its memory, then registers the output signal into its memory as the new internal state

References

- Kosiba, B. "Arabinose-inducible Promoter from Escherichia Coli: Its Cloning from Chromosomal DNA, Identification as the AraFG Promoter and Sequence." *Journal of Molecular Biology* 156.1 (1982): 53-66. Print.
- Lou, Chunbo, Xili Liu, Ming Ni, Yiqi Huang, Qiushi Huang, Longwen Huang, Lingli Jiang, Dan Lu, Mingcong Wang, Chang Liu, Daizhuo Chen, Chongyi Chen, Xiaoyue Chen, Le Yang, Haisu Ma, Jianguo Chen, and Qi Ouyang. "Synthesizing a Novel Genetic Sequential Logic Circuit: a Push-on Push-off Switch." *Molecular Systems Biology* 6 (2010). Print.
- Purnick, Priscilla E. M., and Ron Weiss. "The Second Wave of Synthetic Biology: from Modules to Systems." *Nature Reviews Molecular Cell Biology* 10.6 (2009): 410-22. Print.

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Questions?

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Presented by Sana Wajid

GSLC: Nor gate Module, ON State

